TOPIC 1: INTRODUCTION TO MACHINE LEARNING

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 – Sem 1

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COURSE LEARNING OUTCOMES

At the end of this topic, students should be able to:

a. Define machine learning and why it is important.

b. Discover examples of machine learning applications.

c. Distinguish the types of machine learning systems.

d. Identify main challenges of machine learning.

Topic 1: Introduction to Machine Learning

- Defining machine learning and why is it important.
- Discover examples of machine learning applications.
- Distinguish the types of machine learning systems.
- Identify main challenges of machine learning.

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What is Artificial Intelligence?

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What is Artificial Intelligence?



The goal of AI is to create computer models that exhibit "intelligent behaviors" like humans.

—Boris Katz

Source: https://mitsloan.mit.edu/ideas-made-to-matter/machine-learning-explained

What is Machine Learning?



Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.

—Arthur Samuel, 1959

Source: https://www.ibm.com/history/early-games

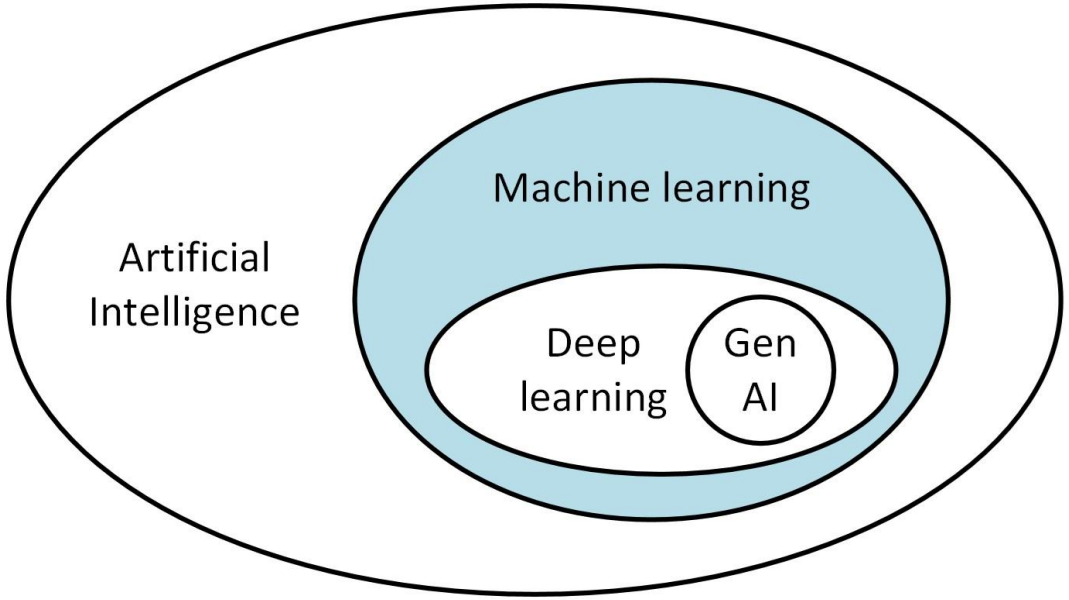
What is Machine Learning?

Generally, we make decisions in the following two ways:

- By using logic and reasoning.
- By using our experience.

Computer make decisions based on data. Experience here means data.

Anytime computer solve a problem or make decision using only data, we are doing machine learning.



Machine learning is a part of artificial intelligence.

History of Machine Learning

1950 - Alan Turing creates the “Turing Test” to determine if a computer has real intelligence.

1952 - Arthur Samuel wrote the first computer learning program.

1956 - John McCarthy introduces the “Artificial Intelligence” term.

1957 - Frank Rosenblatt designed the first neural network for computers (the perceptron).

1966 - ELIZA, the first chatbot is created by Joseph Weizenbaum at MIT.

1967 - The “nearest neighbor” algorithm was written. This allowed computers to perform very basic pattern recognition.

1979 - Students at Stanford University invent the “Stanford Cart”.

1981 - Gerald Dejong introduces the concept of Explanation Based Learning (EBL)

2006 - Geoffrey Hinton coins the term “deep learning”.

2017 - Transformer model. (Attention is all you need paper).

2019 - GPT-2 release.

2020 - GPT-3 release.

2021 - Vision language model.

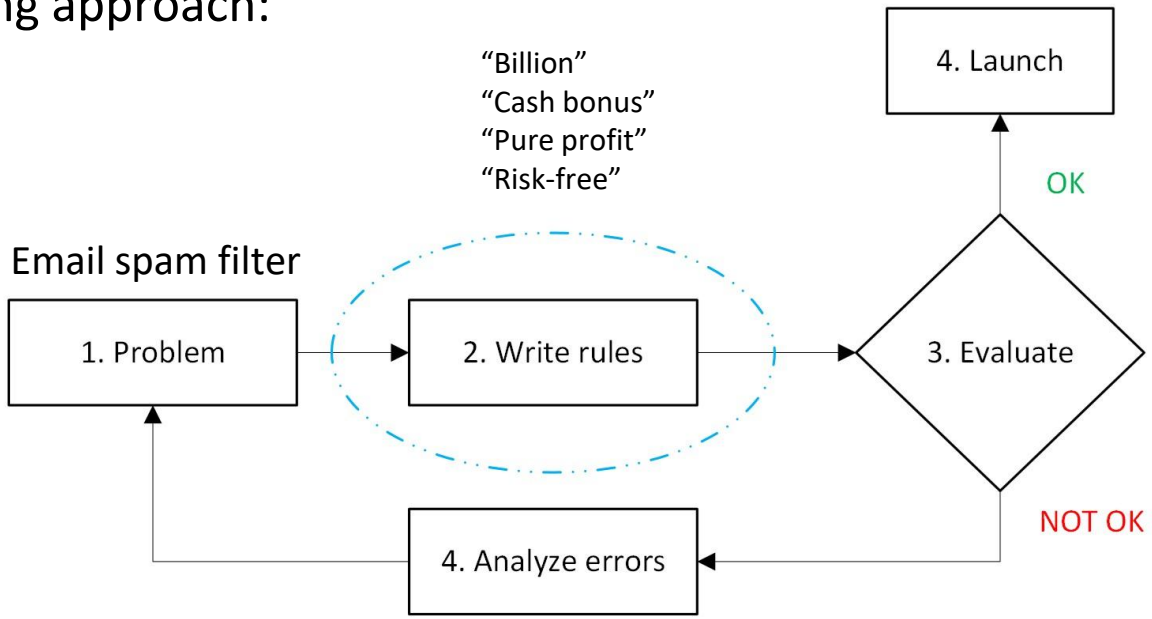
2023 - GPT-4.

2025 - Deepseek, Agentic AI (Manus AI).

Source: https://www.forbes.com/sites/bernardmarr/2016/02/19/a-short-history-of-machine-learning-every-manager-should-read/

Why do we use Machine Learning?

Traditional programming approach:



Source: Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurelien Geron (2023)

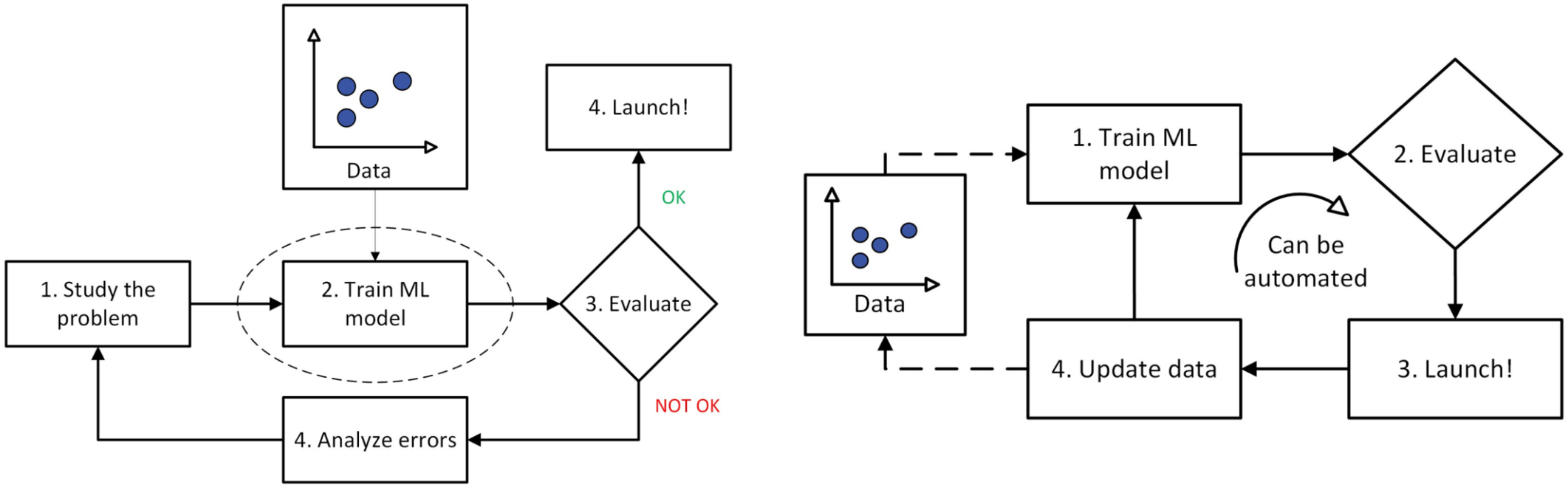
Eventually, if spammers keep working around your spam filter, the program will likely become a long list of complex rules—hard to maintain.

Source: Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurelien Geron (2023)

Why do we use Machine Learning?

Machine learning approach:

- ML automatically learns which words and phrases are good predictors of spam.
- It is also adaptive to change.



Source: Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurelien Geron (2023)

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Why do we use machine learning?

Machine learning is great for:

1. Problems with solutions that require a lot of fine-tuning or long lists of rules. E.g., biomedical imaging, fraud detection.
2. Complex problems for which using a traditional approach yields no optimal solution E.g., autonomous driving.
3. Fluctuating environments. E.g., weather patterns, real-time traffic conditions
4. Getting insights about complex problems and large amounts of data. E.g., financial market analysis, sentiment analysis on social media.

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Models and Algorithms

Algorithm A procedure, or a set of steps, used to solve a problem or perform a computation.

Model A set of rules that represent our data and can be used to make predictions.

- An algorithm is what we use to build the model.
- A model is what we use to make predictions.
- Those two definitions are easy to confuse and are often interchanged.

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Examples of ML Applications

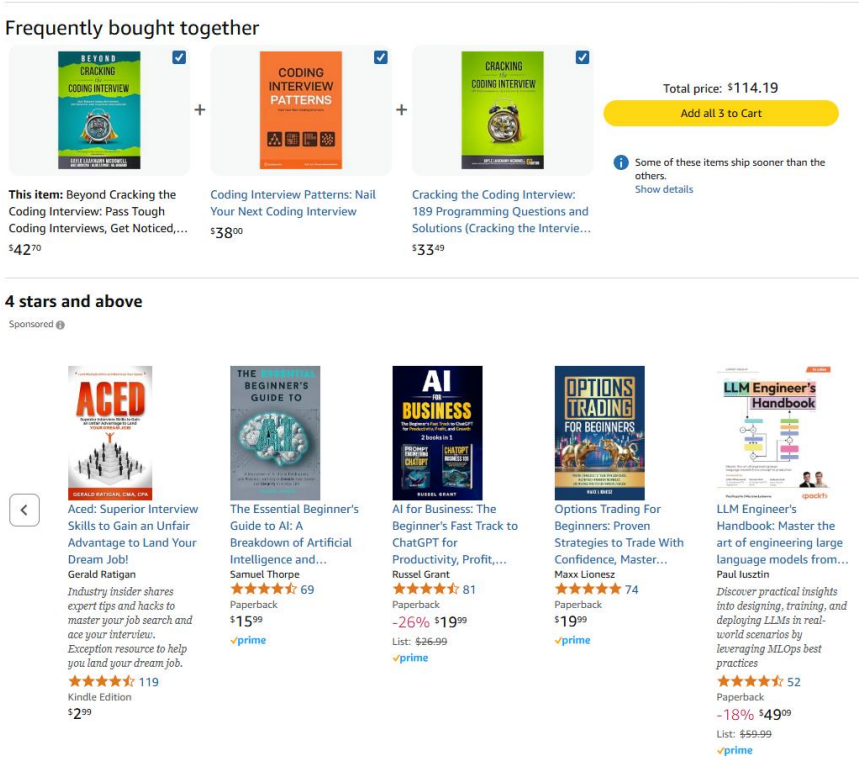
Machine learning use cases:

- Marketing and sales
- Customer service
- Personal and voice assistants
- Filtering email
- Cybersecurity
- Healthcare
- Transportation
- Social media

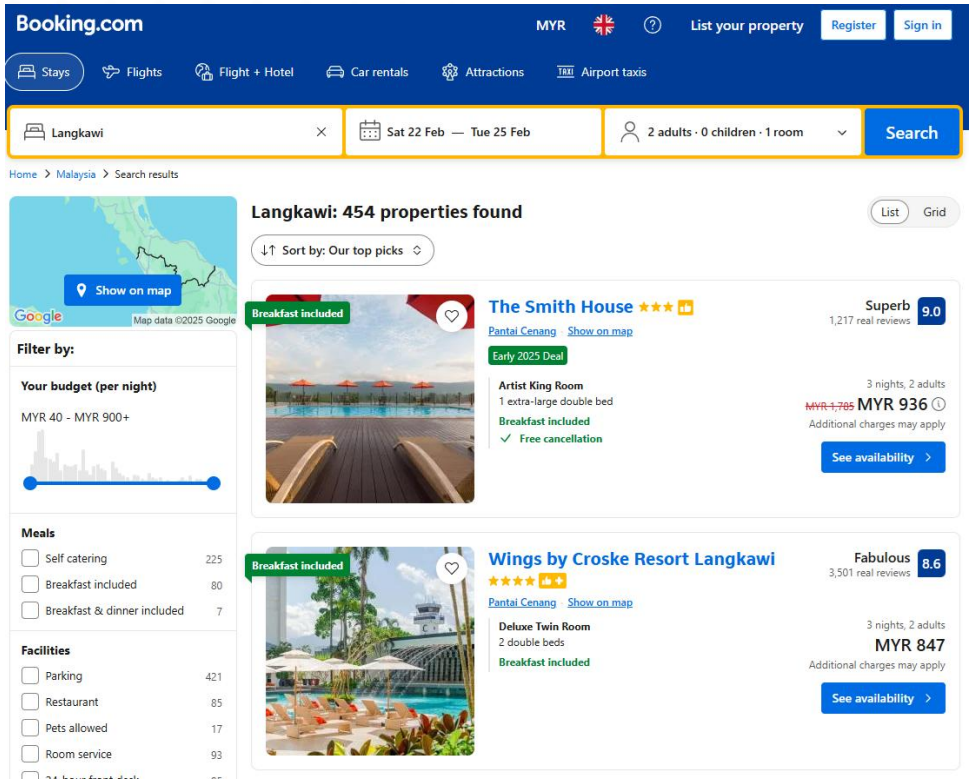
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Marketing and sales

Amazon make recommendations based on user's taste, browsing and shopping cart history.

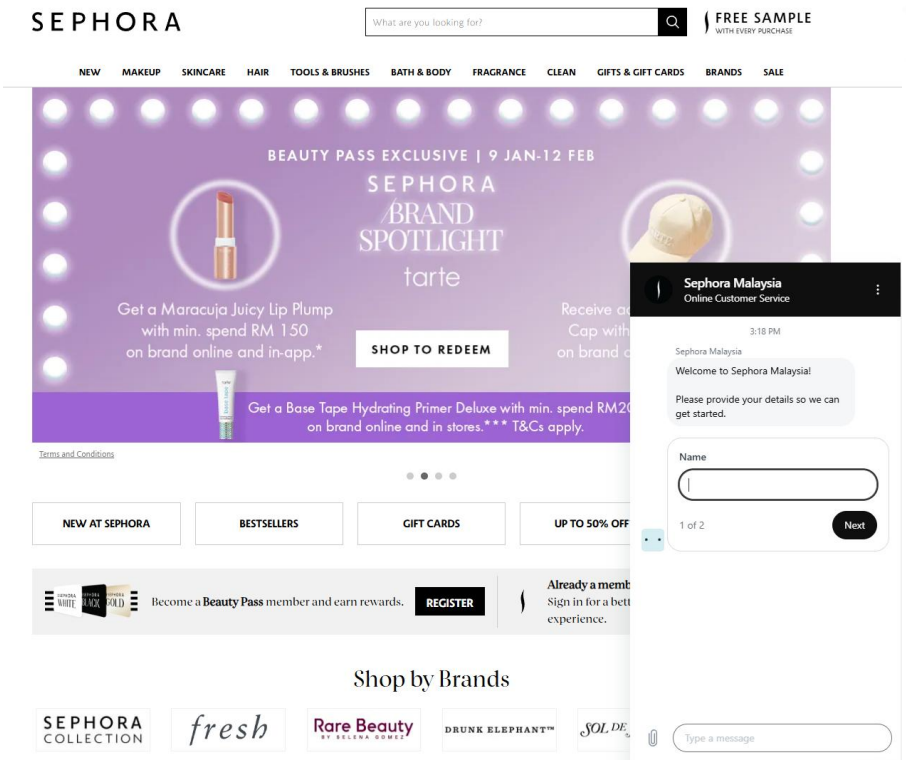


Booking.com adjust prices according to customer's booking patterns.

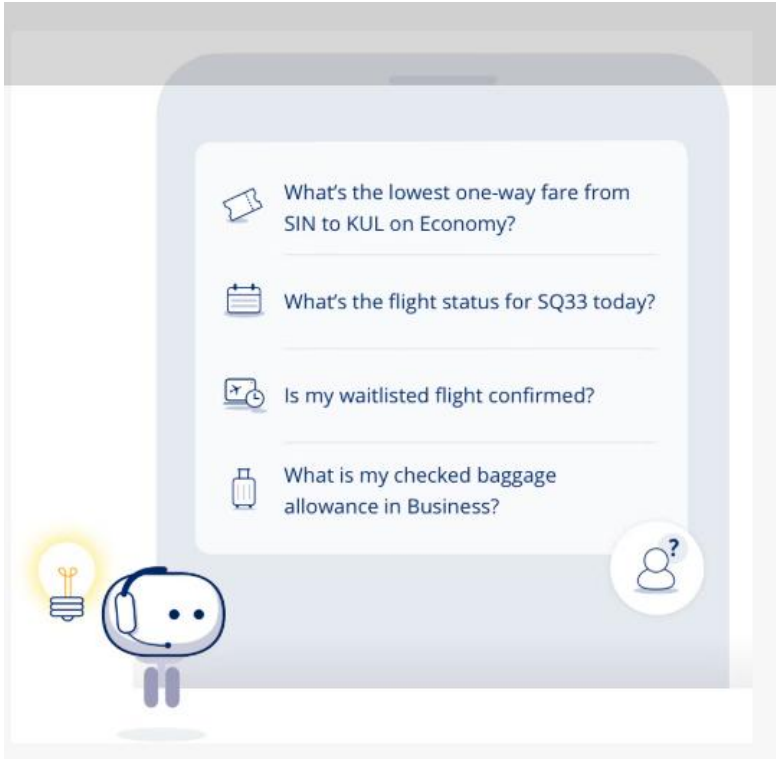


Customer service

Cosmetic giant, Sephora using a chatbot to provide one-click customer service.



Kris, the Singapore Airlines chatbot can help customer with common travel topics.



Personal & voice assistant

Siri on iPhone can answer questions and perform tasks.

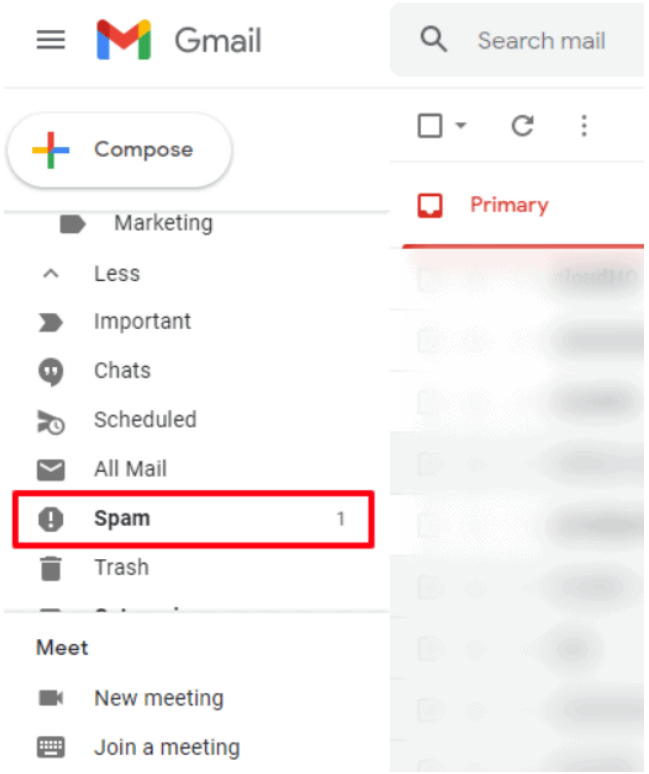


Alexa in Amazon Echo can control music and manage to-do list.

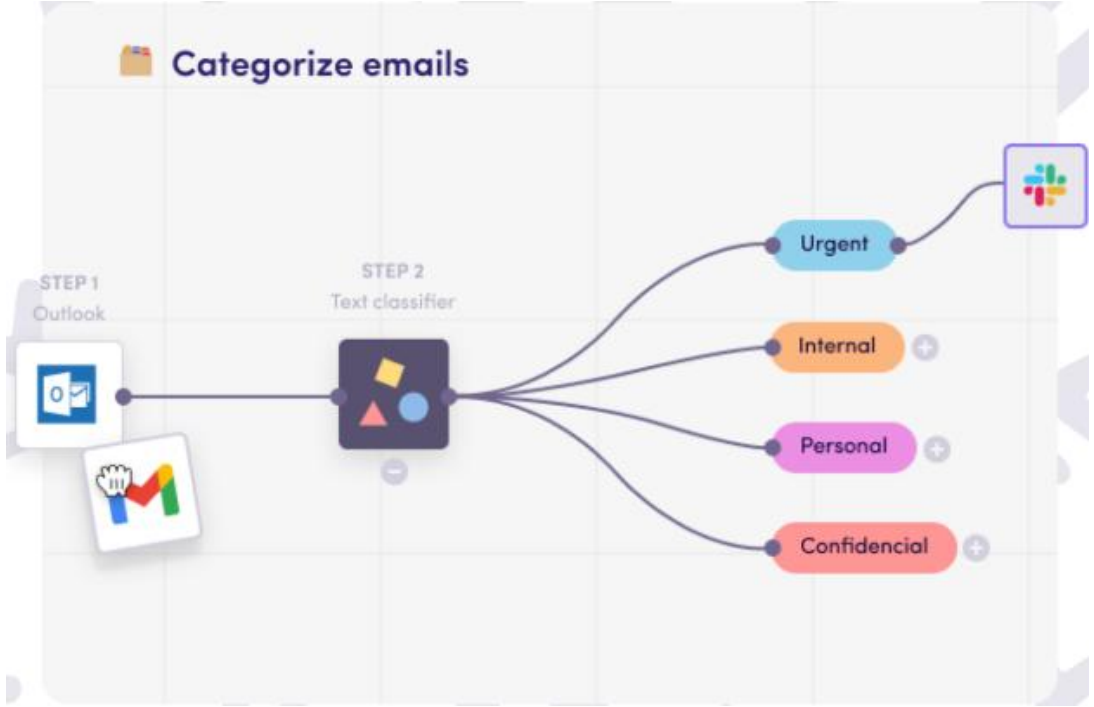


Filtering email

ML algorithms in Google's Gmail can detect and reroute spam to spam folder.

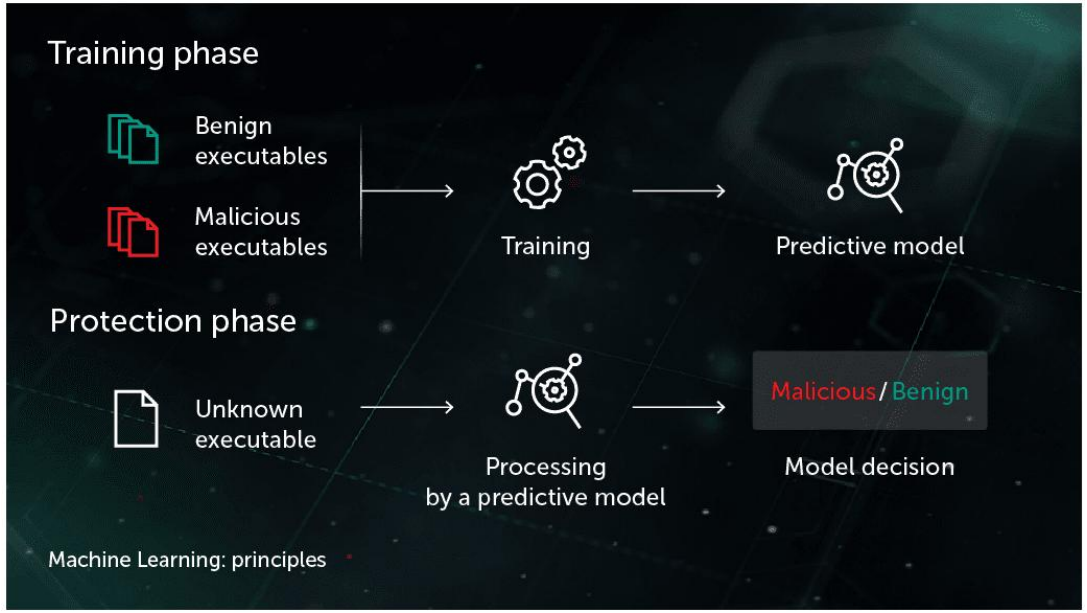


Levity use ML to classify business emails based on sentiment, topic, urgency and team.



Cybersecurity

Antivirus program like Kaspersky use ML to detect and block malware.

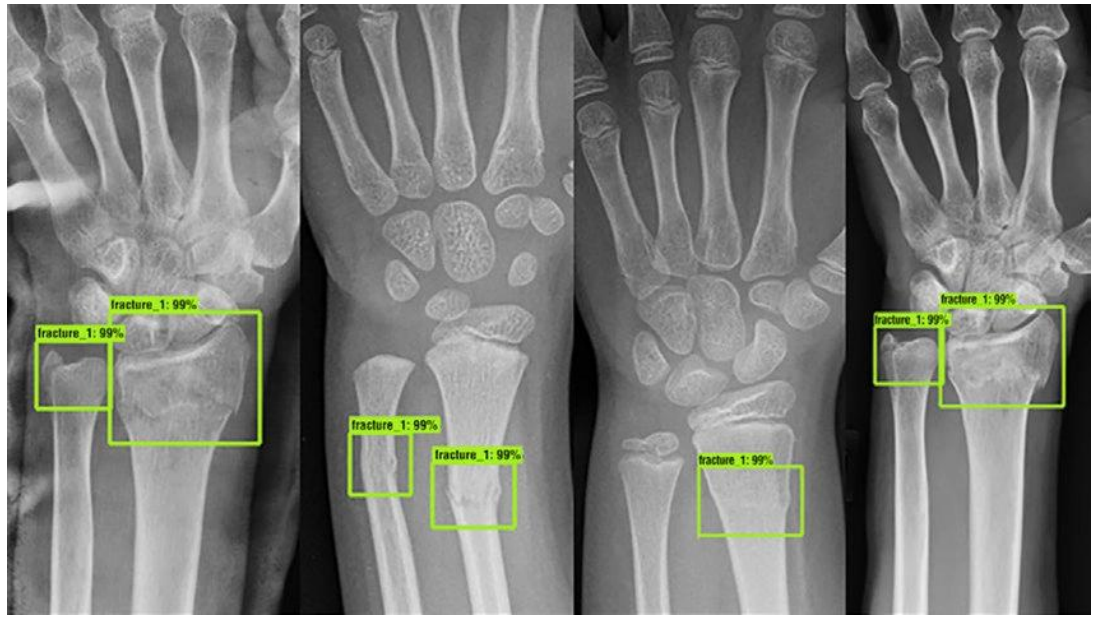


Mastercard use ML to identify patterns for fraud detection during payments.



Healthcare

ML can detect bone fractures that are difficult to see with the human eye.

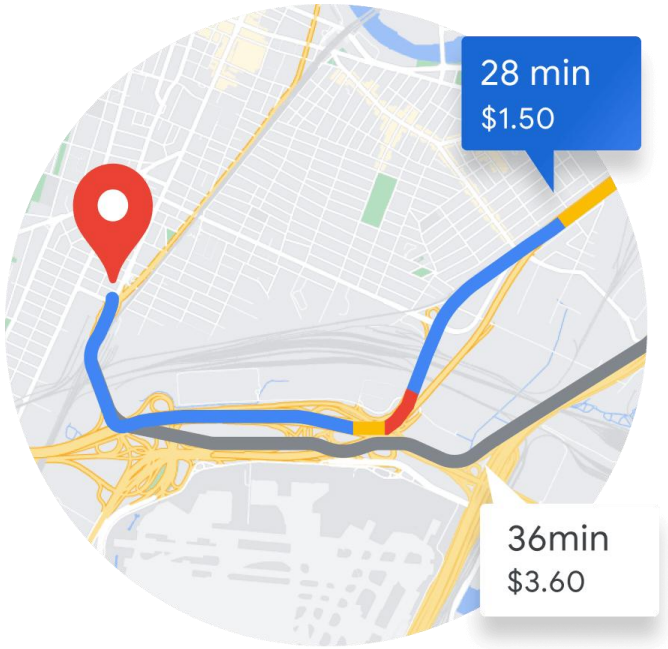


ML can help to make clinical drug development faster.

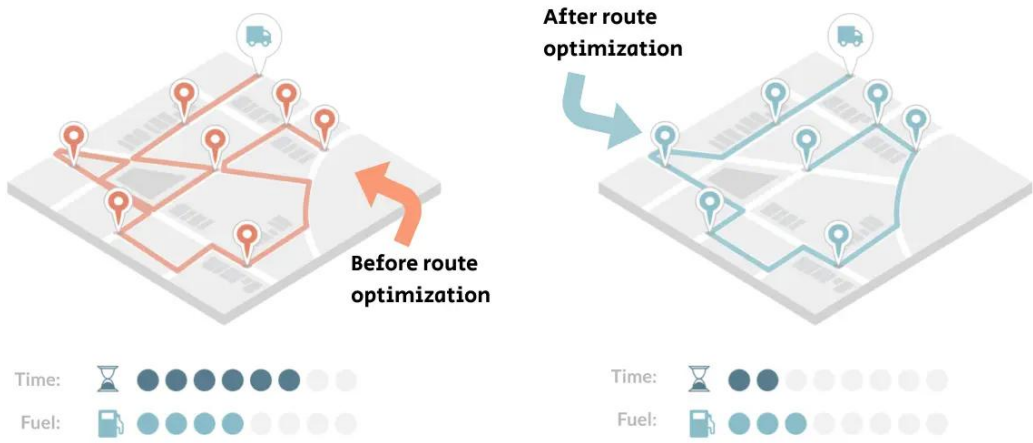


Transportation

Google Maps uses ML algorithms to check the current traffic conditions.

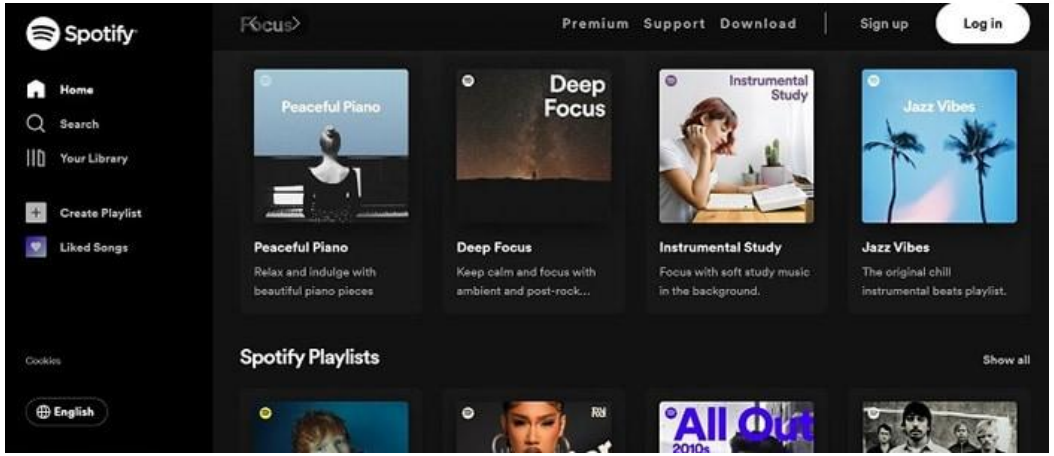


Routific offers route optimization software for deliver businesses.

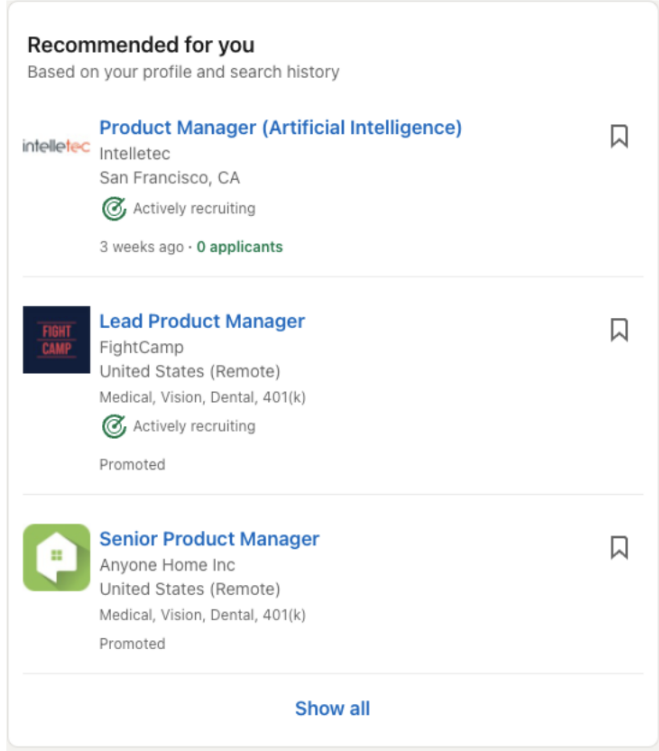


Social media

Spotify uses ML models to generate its song recommendations.



LinkedIn uses ML to make employment recommendations.



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Can you list other examples of ML applications?

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What is a machine learning model?

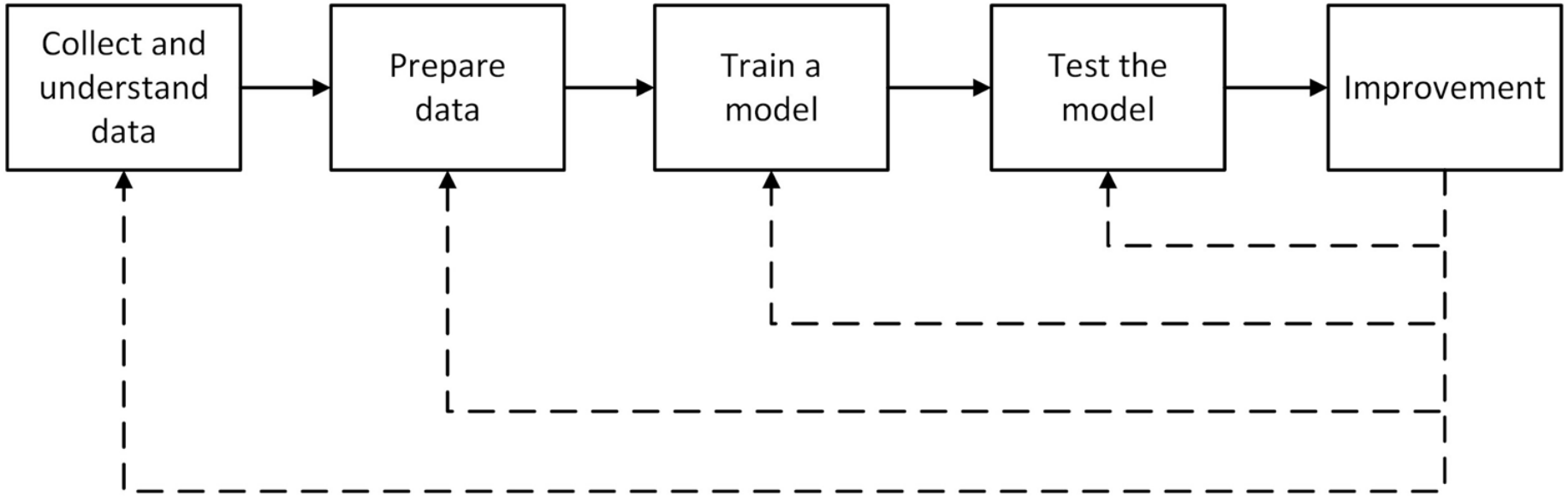
Two types of problems dominate machine learning are regression and classification.

- Regression: estimates the relationship between a target outcome label and one or more feature variables to predict a continuous numeric value.
- E.g., Linear regression, ridge regression, decision trees, random forest, k-nearest neighbour, and neural network regression.

- Classification: identifies category an item belongs based on labelled examples of known items.
- E.g., Logistic regression, Naïve Bayes, Decision trees, Random forest, K-nearest neighbour, and support vector machine.

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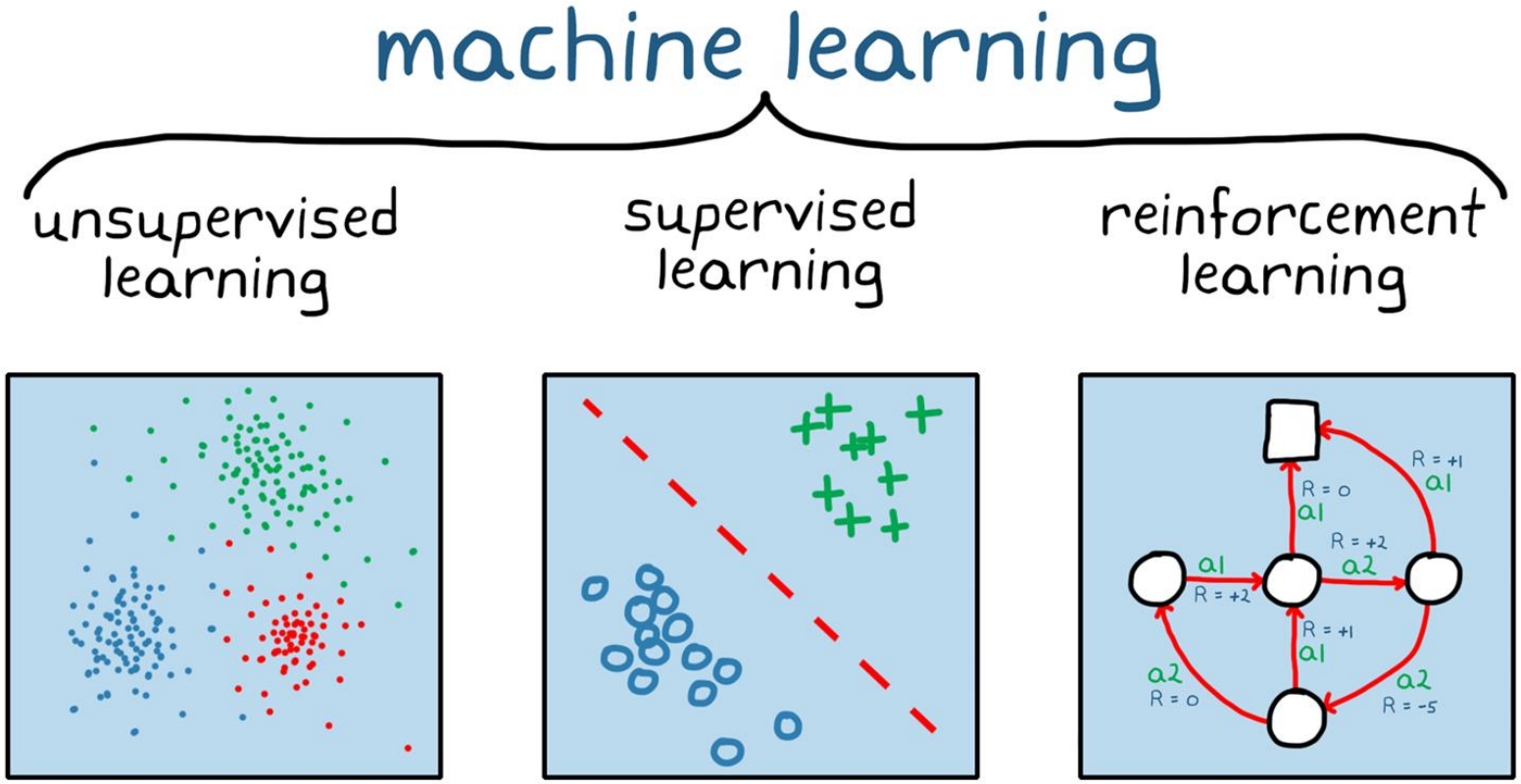
Machine learning lifecycle



Source: Grokking Artificial Intelligence Algorithms by Rishal Hurbans (2020)

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Types of Machine Learning Systems



Source: MathWorks

Types of Machine Learning Systems

ML systems can be categorized based on the following criteria:

- How are they supervised during training? (supervised, unsupervised, semi-supervised, self-supervised, and others)
- Can the system learn incrementally from incoming data? (batch vs online learning)
- How they generalize? (instance-based vs model-based learning)

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Supervised learning

- Supervised learning: The algorithm's training uses labelled data or answer key.

- E.g., Iris flower classification, animal classification.



Iris Versicolor **Iris Setosa** **Iris Virginica**

Source: https://github.com/leechee/Iris-Flower-Classification-Tutorial

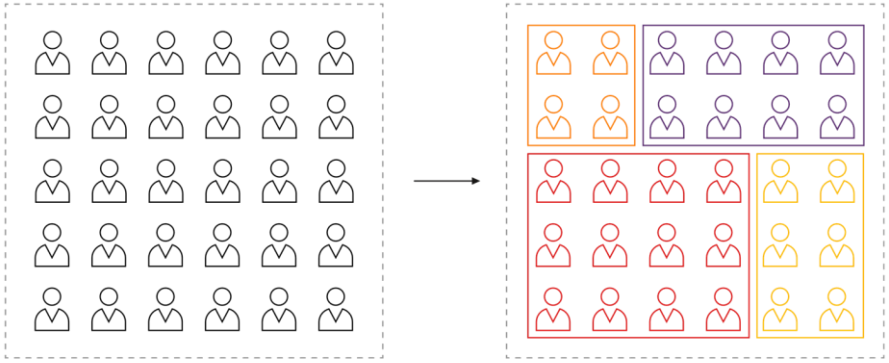


Source: https://www.kaggle.com/datasets/bhavikjikadara/dog-and-cat-classification-dataset

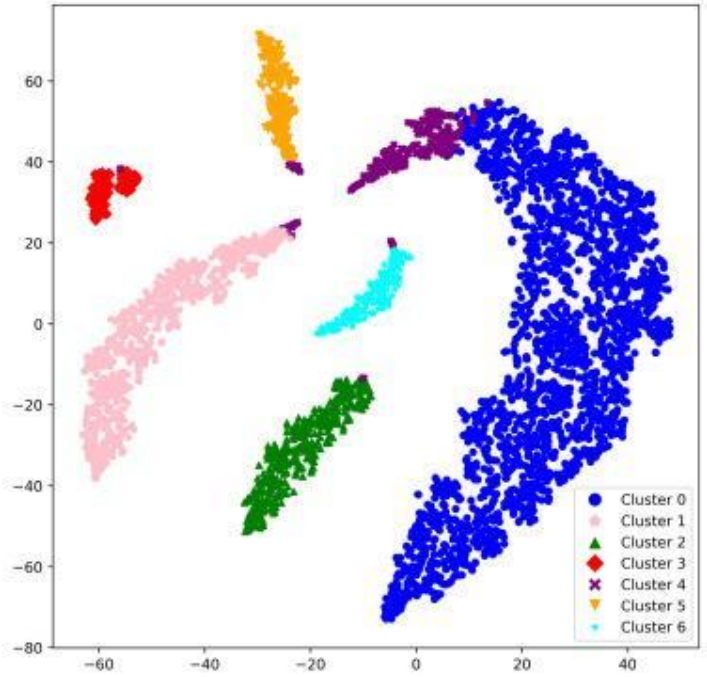
Unsupervised learning

- Unsupervised learning: The algorithm's training use unlabelled data to find unknown patterns.

- E.g. Customer segmentation, sentiment analysis, residential electricity load profiles



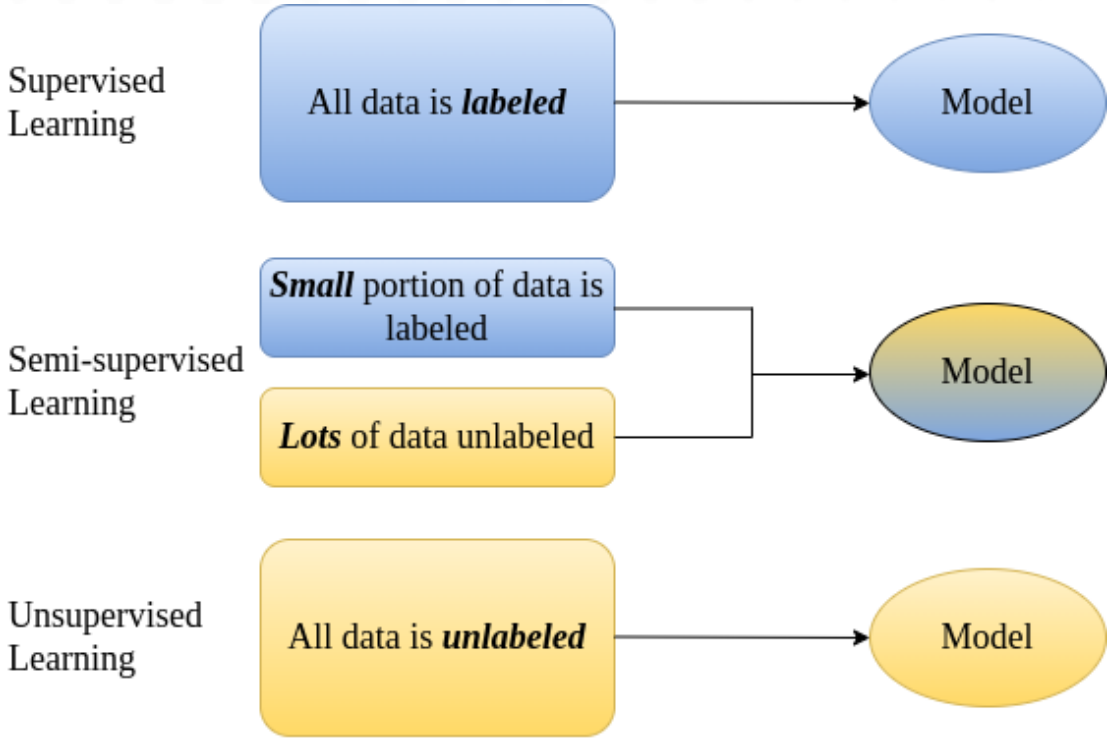
Source: https://www.univio.com/blog/machine-learning-and-customer-segmentation-meet-the-perfect-couple/



Source: Michalakopoulos, V., Sarmas, E., Papias, I., Skaloumpakas, P., Marinakis, V., & Doukas, H. (2024). A machine learning-based framework for clustering residential electricity load profiles to enhance demand response programs. *Applied Energy*, *361*, 122943. https://doi.org/10.1016/j.apenergy.2024.122943

Semi-supervised learning

- Semi-supervised learning: The algorithm's training uses both unlabelled and labelled data.
- Labelling large dataset for supervised learning can be expensive and time-consuming task.
- E.g. Text classification.

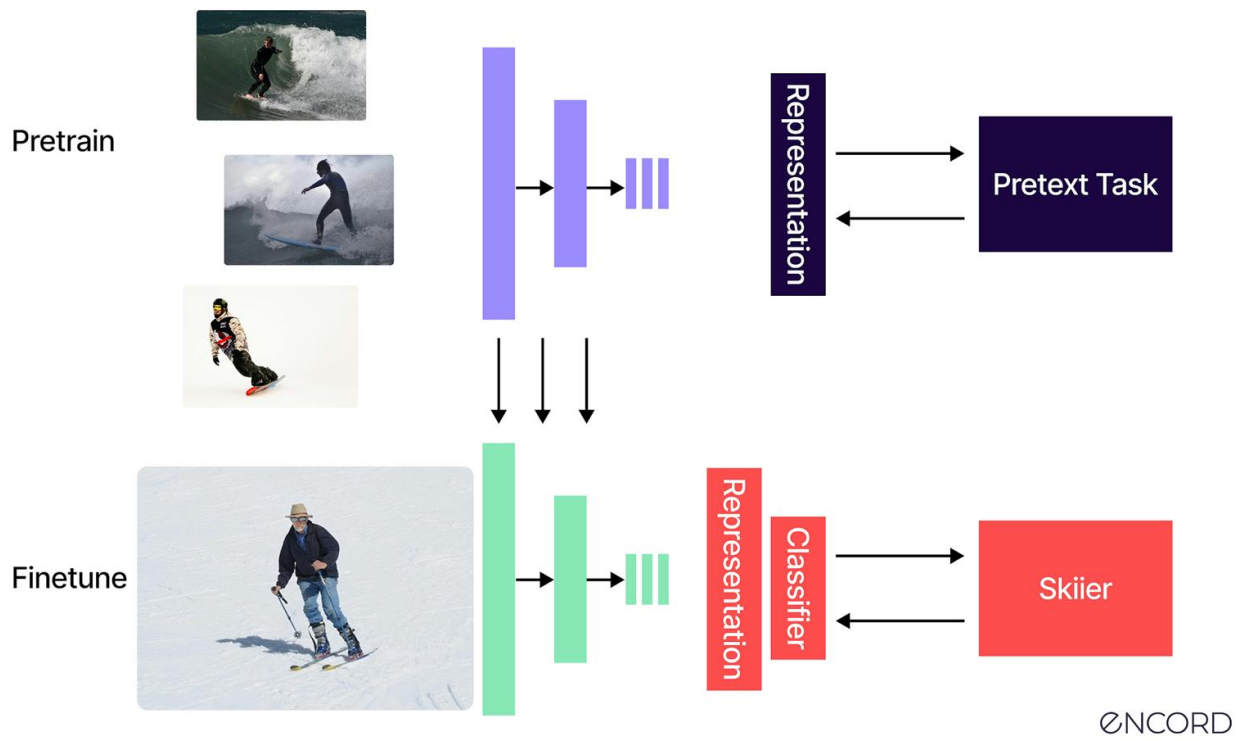


Source: https://blog.roboflow.com/what-is-semi-supervised-learning/

Self-supervised learning

- Self-supervised learning is a machine learning technique that uses unsupervised learning for tasks that conventionally require supervised learning. It encourage a model to train from unlabeled data.

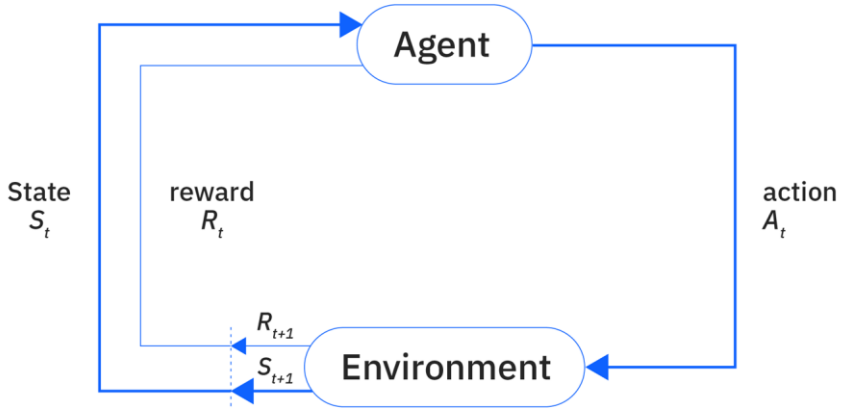
- E.g., Computer vision and natural language processing.



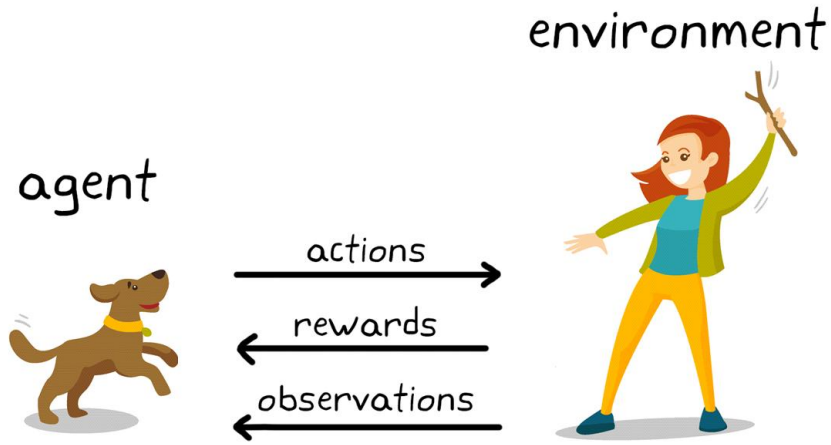
Source: https://encord.com/blog/self-supervised-learning/

Reinforcement learning

- Reinforcement learning: The algorithm or agent learns by interacting with its environment and getting positive or negative reward.
- E.g., Games, Robotics, Self-driving cars



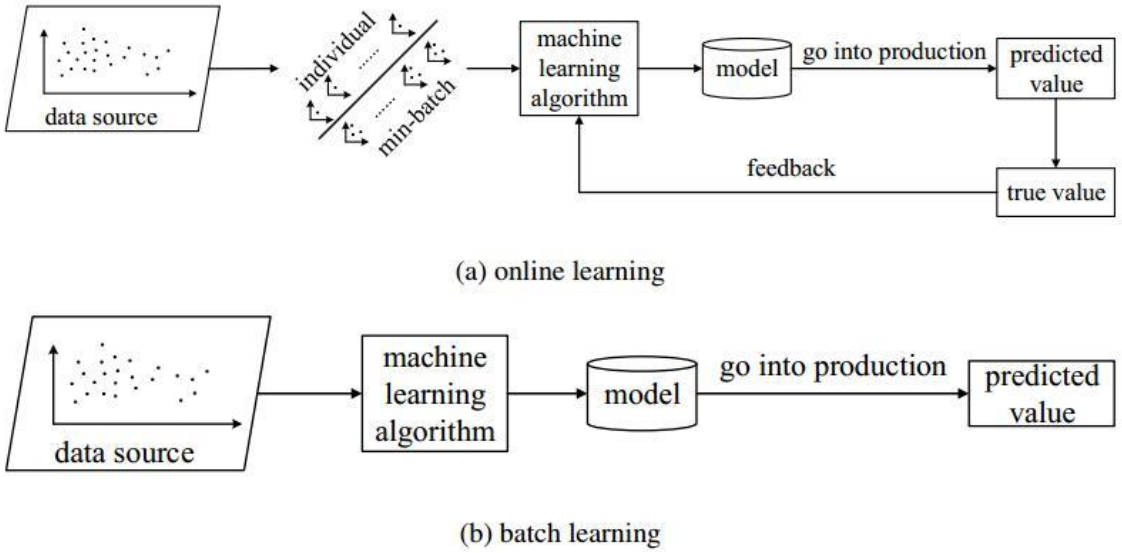
Source: https://www.ibm.com/think/topics/reinforcement-learning



Source: https://www.mathworks.com/discovery/reinforcement-learning.html

Batch learning vs online learning

- Batch learning, also known as offline learning is incapable of learning incrementally.
- Online learning is when the ML model incrementally trained by feeding it data instances sequentially, either individually or in small groups (mini-batches).

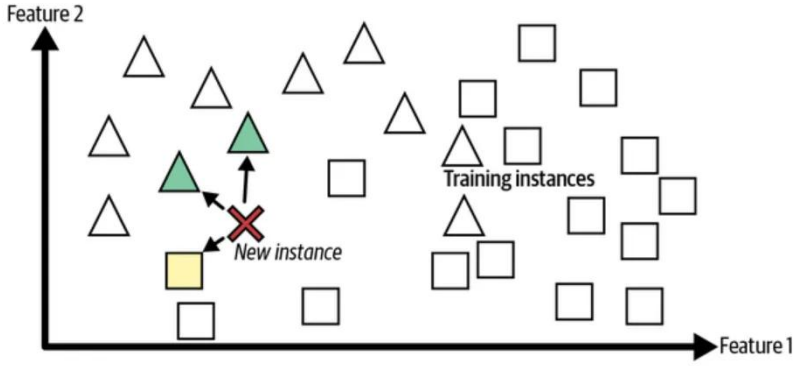


Source: Xue, W., Wan, P., Li, Q., Zhong, P., Yu, G., & Tao, T. (2020). An online conjugate gradient algorithm for large-scale data analysis in machine learning. *AIMS Mathematics*, *6*(2), 1515–1537. https://doi.org/10.3934/math.2021092

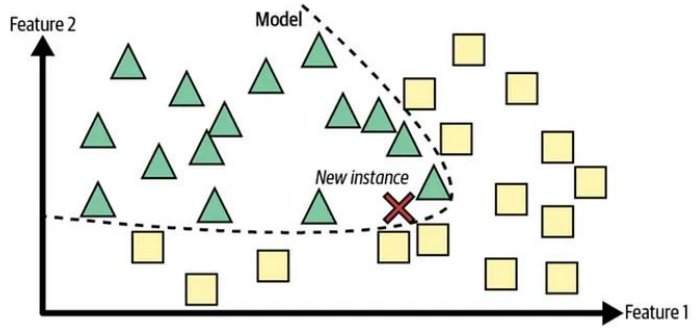
Instance-based learning vs model-based learning

- The instance-based learning learns from examples and stores them in memory. The model can then make predictions by comparing new input data to the stored examples.

- The model-based learning learns the underlying relationships and patterns in the data by creating a mathematical representation.



Instance-based learning



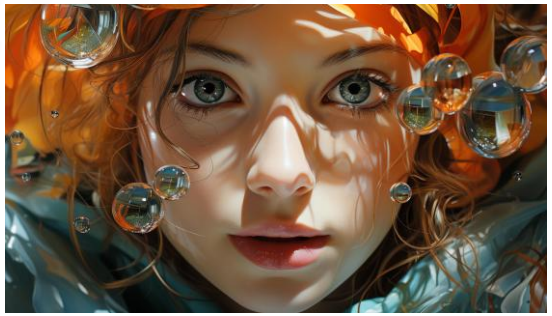
Model-based learning

Source: https://medium.com/@pp1222001/model-based-vs-instance-based-learning-understanding-the-differences-with-examples-1545c9c3a056

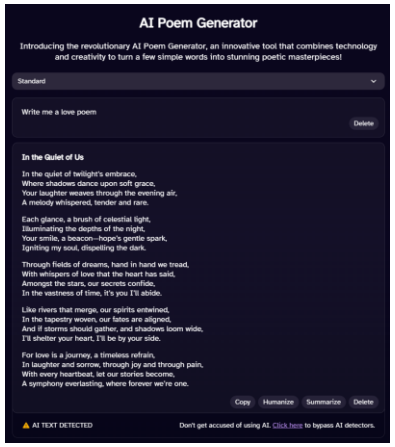
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Generative machine learning

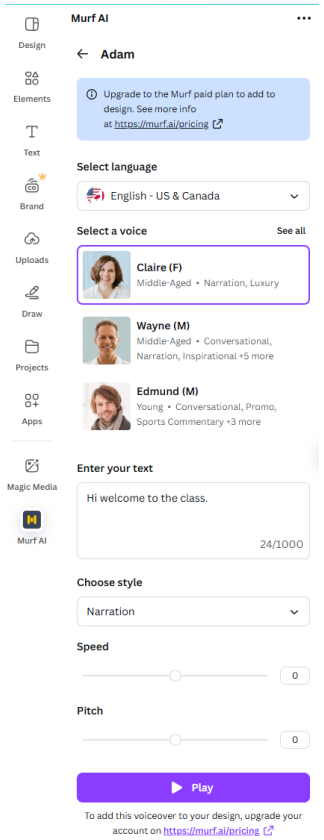
- The field of generative learning consists of models that, given a dataset can output new data points that look like samples from the original dataset.
- Generative algorithms have been able to create realistic images, paintings, video, music, stories, poetry, etc.
- Examples of generative algorithms are: generative adversarial networks (GANs), variational autoencoders, and restricted Boltzmann machines (RBMs).



Ai Generated Stock photos by Vecteezy



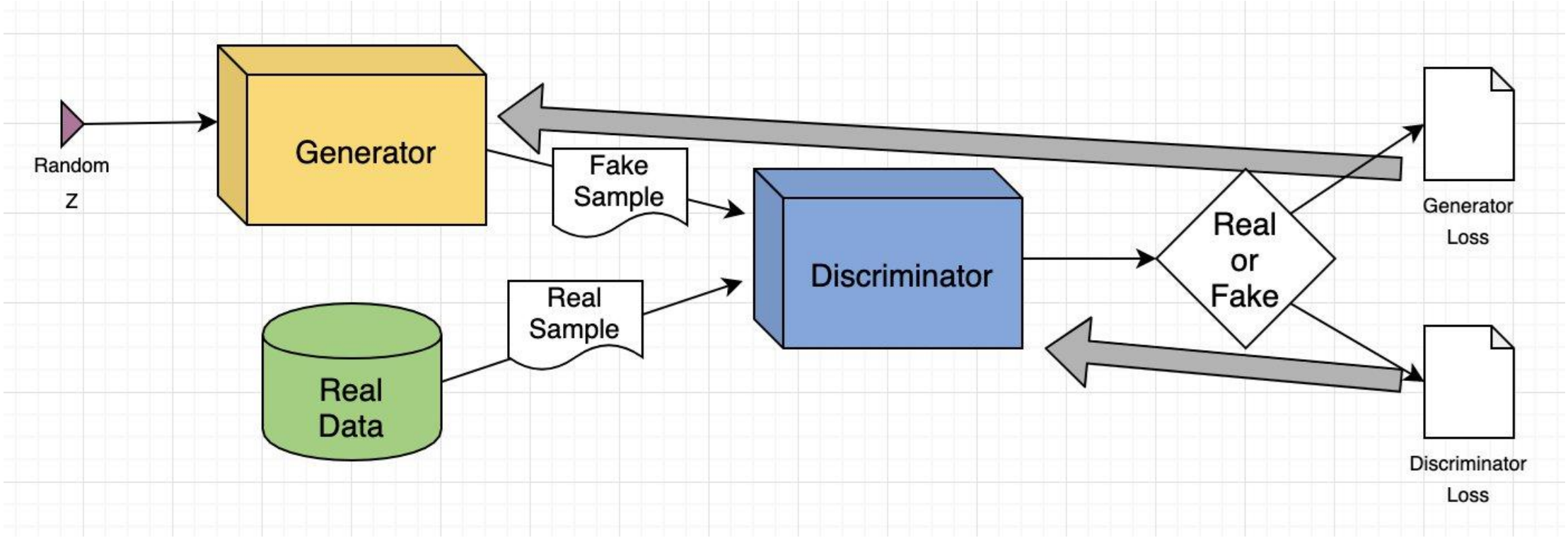
AI Poem Generator



Canva Murf AI

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Generative adversarial network



Source: Amazon AWS

A generative adversarial network system comprises two deep neural networks—the *generator network* and the *discriminator network*. Both networks train in an adversarial game, where one tries to generate new data and the other attempts to predict if the output is fake or real data.

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- Defining machine learning and why is it important.
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- Identify main challenges of machine learning.

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Challenges of Machine Learning

a. Insufficient quantity of training data
b. Non-representative training data
c. Poor-quality data
d. Irrelevant features
e. Overfitting the training data
f. Underfitting the training data

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Insufficient quantity of training data

- Human ability to learn is different from machine. For example, a toddler can recognize apples with few examples only.

- Complex problems such as image or speech recognition may need millions of examples.

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Non-representative training data

- Non-representative training data occurs when the training dataset does not accurately reflect the real-world distribution of data. (sampling bias)

- E.g., COMPAS software used by several states in the US to help predict whether a defendant will recidivate in the next two years if the person is released. The software is found to be discriminatory against African-Americans.

Poor-quality data

Training data can be full of errors, outliers, and noise due to poor data quality measurements.

Examples to clean up training data:

- Discard outlier instances or try to fix the errors manually.
- Fill in the missing values.

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Irrelevant features

- Garbage in, garbage out (GIGO).
- The success of machine learning model is high if the training data contains enough relevant features.
- Feature engineering involves feature selection, feature extraction, or creating new features.

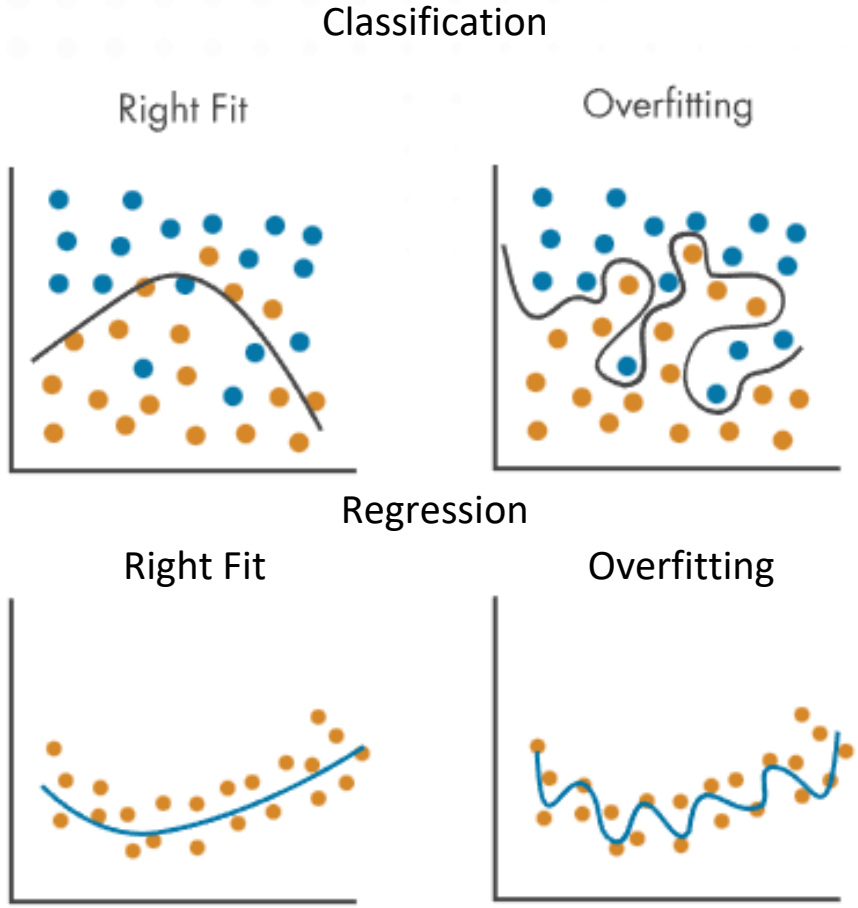
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Overfitting the training data

Overfitting means the model performs well on the training data, but it does not generalize well.

Avoid overfitting by:

- Data cleaning to remove noise.
- Apply regularization techniques.



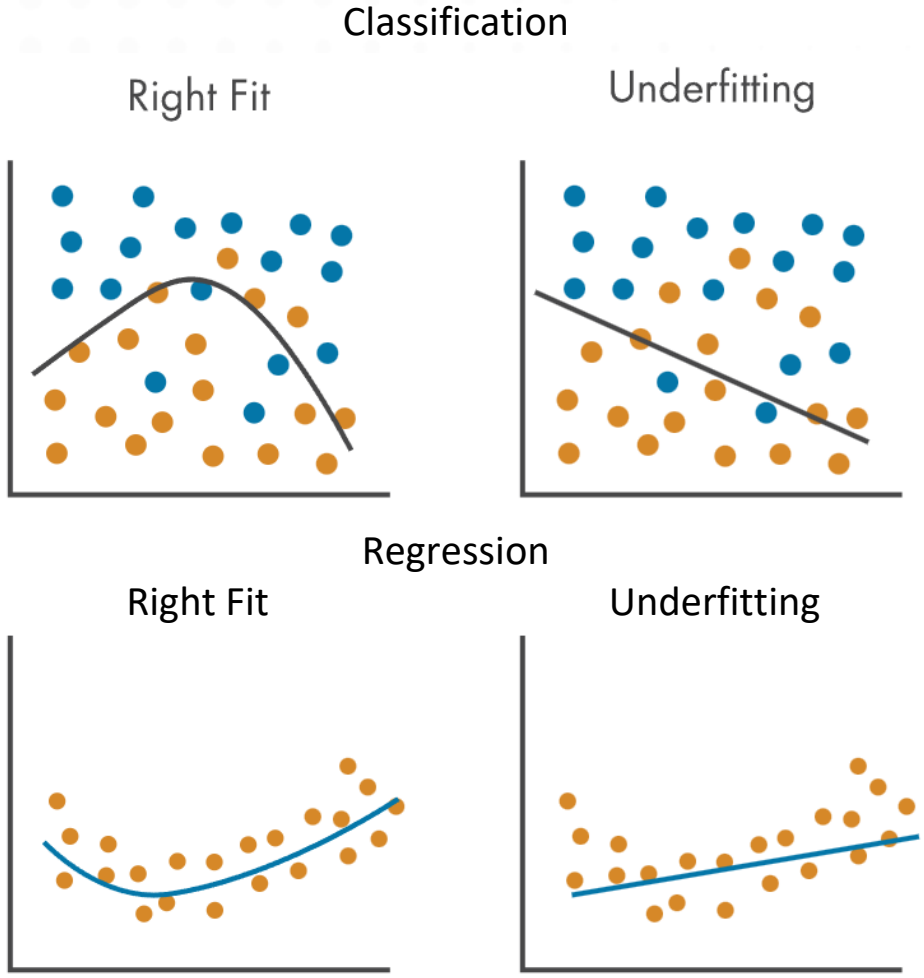
Source: https://www.mathworks.com/discovery/overfitting.html

Underfitting the training data

Underfitting occurs when your model is too simple to learn the structure of the data.

Options to fix underfitting:

- Choose other models with more parameters.
- Feed better features.
- Reduce the constraints. (adjust the regularization hyperparameter).



Source: https://www.mathworks.com/discovery/overfitting.html

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THANK YOU

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